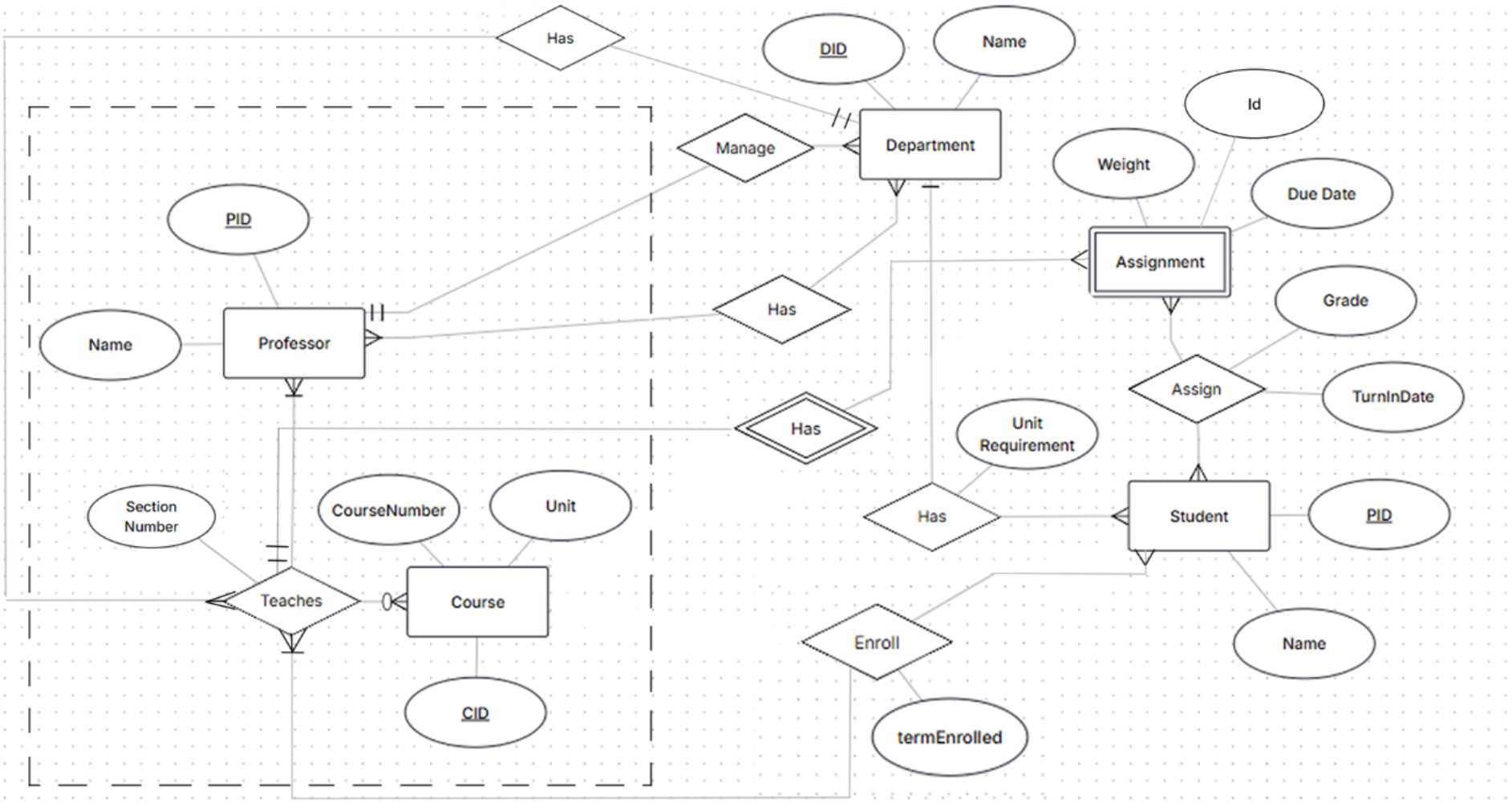
# CS166 Grade Database - ER Diagram

The next page lists the notes used for the SQL schema.



# Schema notes

**Student ID**

Student uses SID. The PID label beside Student in the diagram is a typo.

**Section key**

K = (pid, cid, sectionnumber, term). The term separates sections taught in different quarters. dept depends on K and is not part of the key.

**Assignment and enrollment keys**

Assignment: (K, id). Enroll: (K, sid). Assign: (K, id, sid). Each result must match both an assignment and an enrollment in that section.

**Term**

termEnrolled is stored as the section term in K. It is included in the dependent keys.

**Departments**

Each department has one manager. Managers and section professors must work in that department through WorksIn.

**Adding records**

A section can start without assignments or enrollments. An assignment can start without grades. Each child row still needs its parent record.

**Grades**

Grades and weights are percentages. Assignment IDs are the labels. NULL means the grade or date is unknown; a grade of 0 counts as a recorded grade.

**Weight total**

Assignments must total 100 before finalizing grades. Draft schemes can be incomplete. Final grades are calculated only when the total is 100 and every assignment is graded.